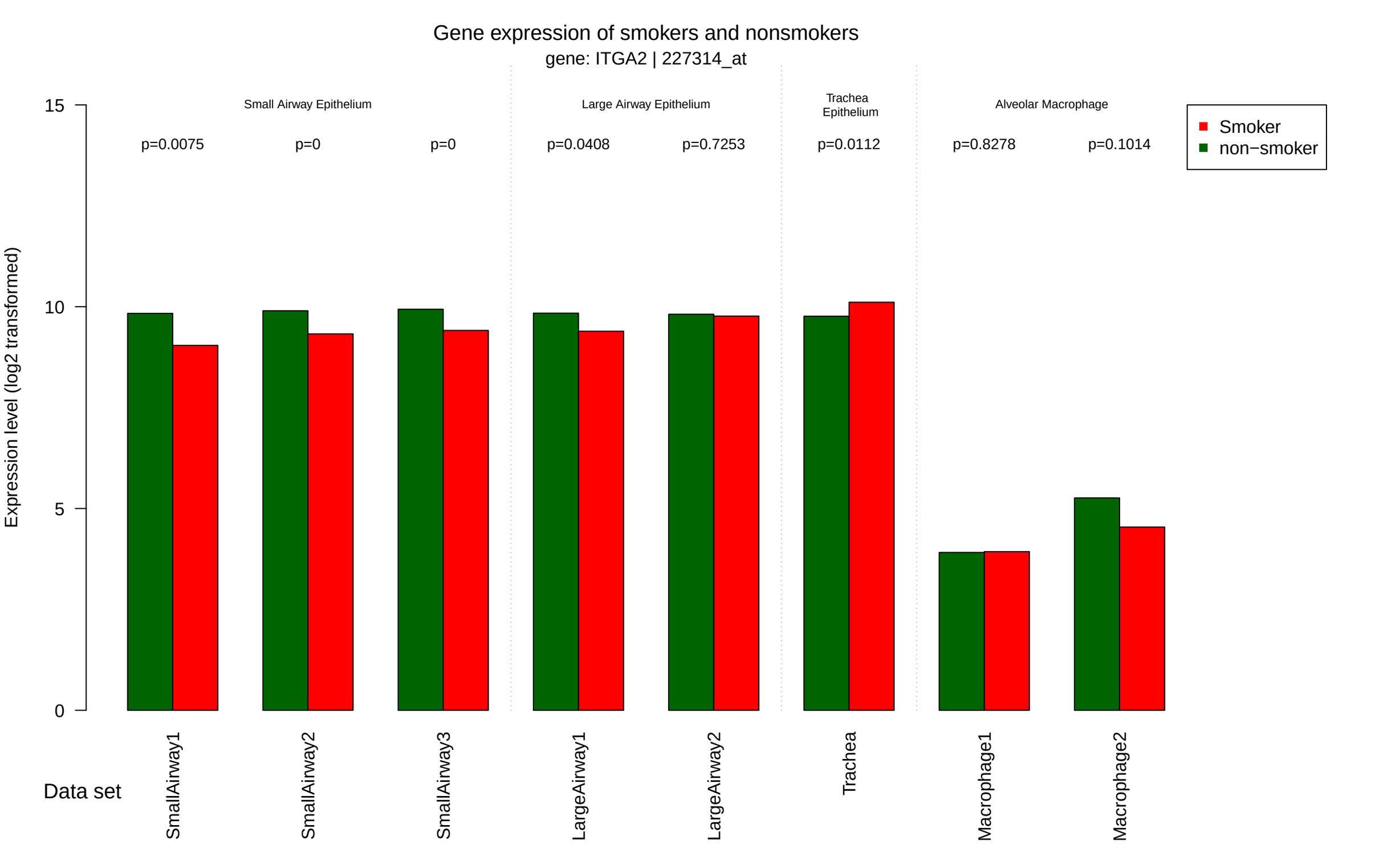
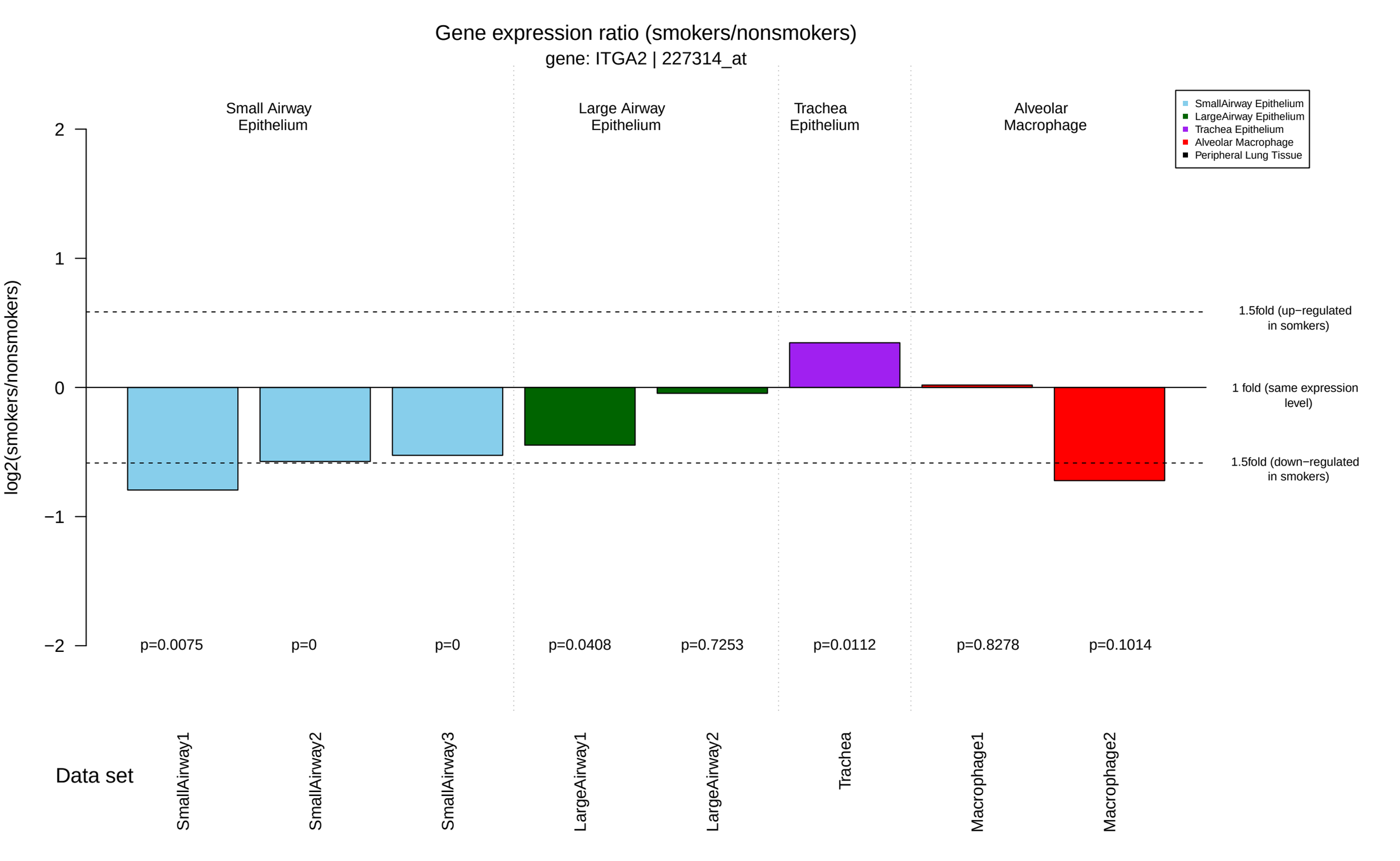
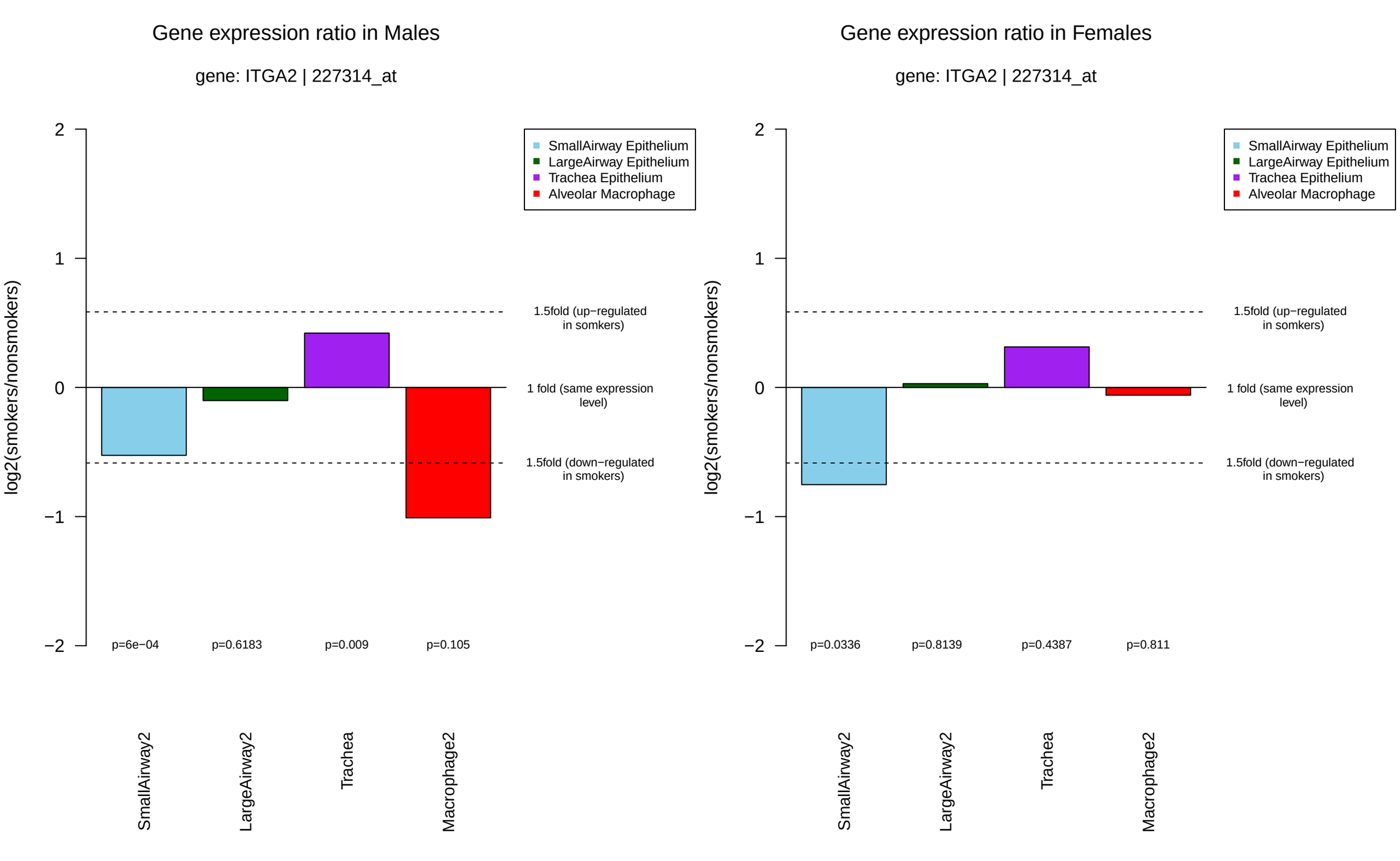
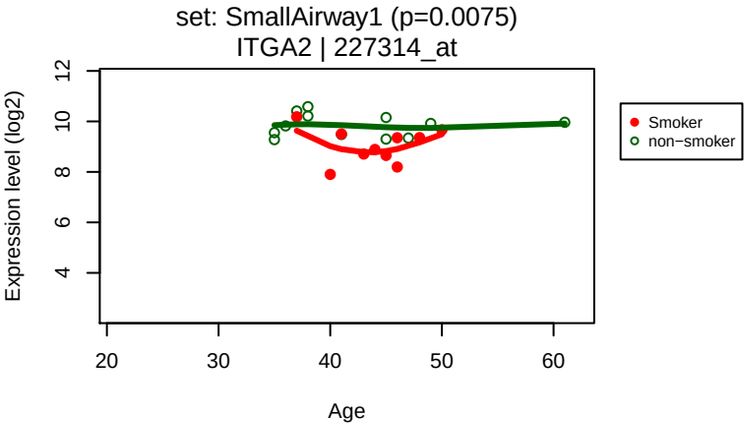
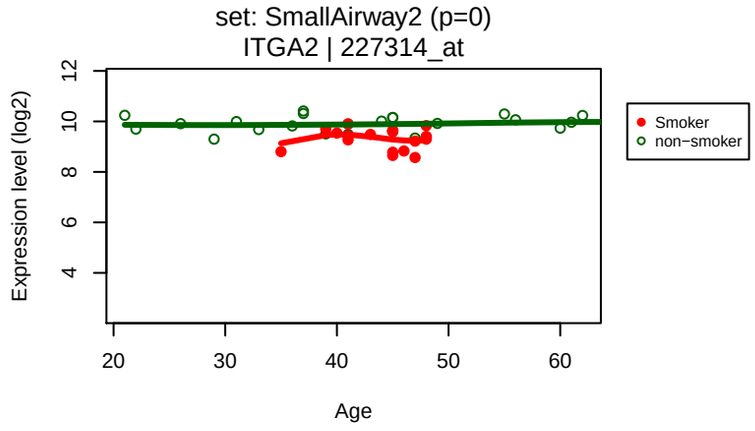
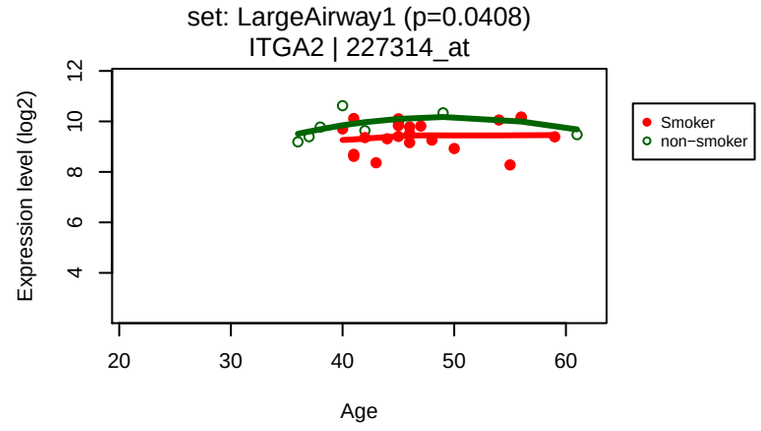

Gene expression of smokers and nonsmokers
gene: ITGA2 | 227314_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0075
p=0
p=0
p=0.0408
p=0.7253
p=0.0112
p=0.8278
p=0.1014
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

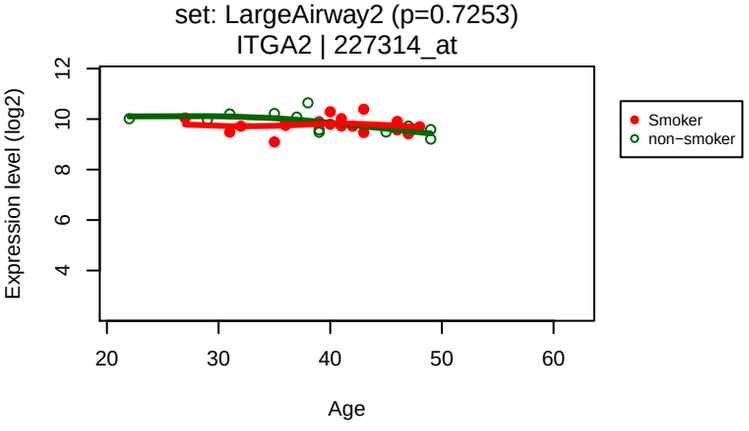

Gene expression ratio (smokers/nonsmokers)
gene: ITGA2 | 227314_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0075
p=0
p=0
p=0.0408
p=0.7253
p=0.0112
p=0.8278
p=0.1014
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

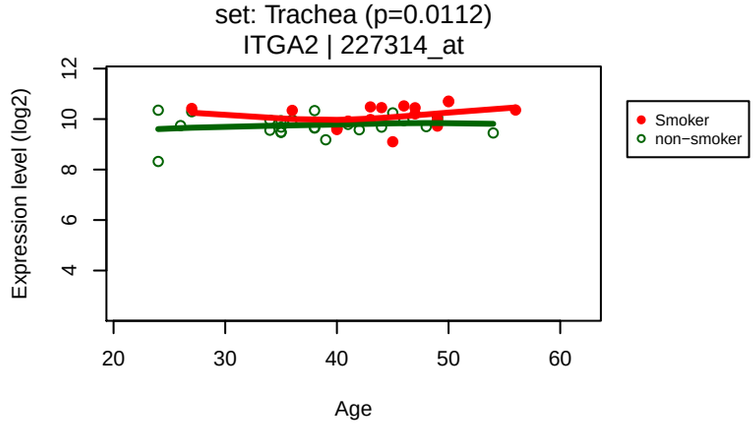

Gene expression ratio in Males
gene: ITGA2 | 227314_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=6e−04
p=0.6183
p=0.009
p=0.105
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ITGA2 | 227314_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0336
p=0.8139
p=0.4387
p=0.811
SmallAirway2
LargeAirway2
Trachea
Macrophage2

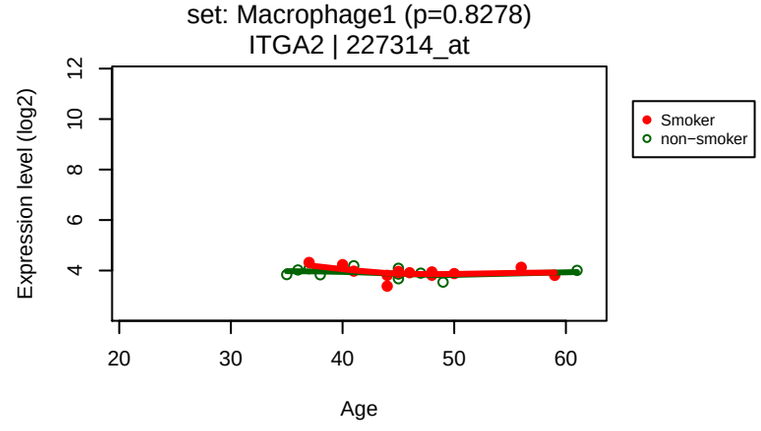

set: SmallAirway1 (p=0.0075)
ITGA2 | 227314_at
Expression level (log2)
Age
Smoker
non−smoker

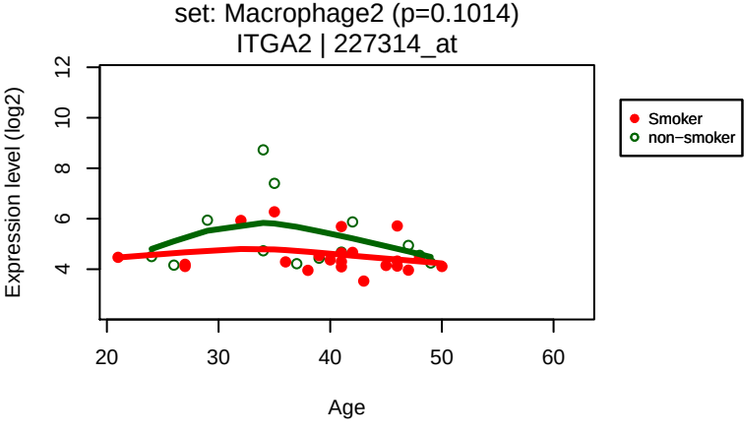

set: SmallAirway2 (p=0)
ITGA2 | 227314_at
Expression level (log2)
Age
Smoker
non−smoker

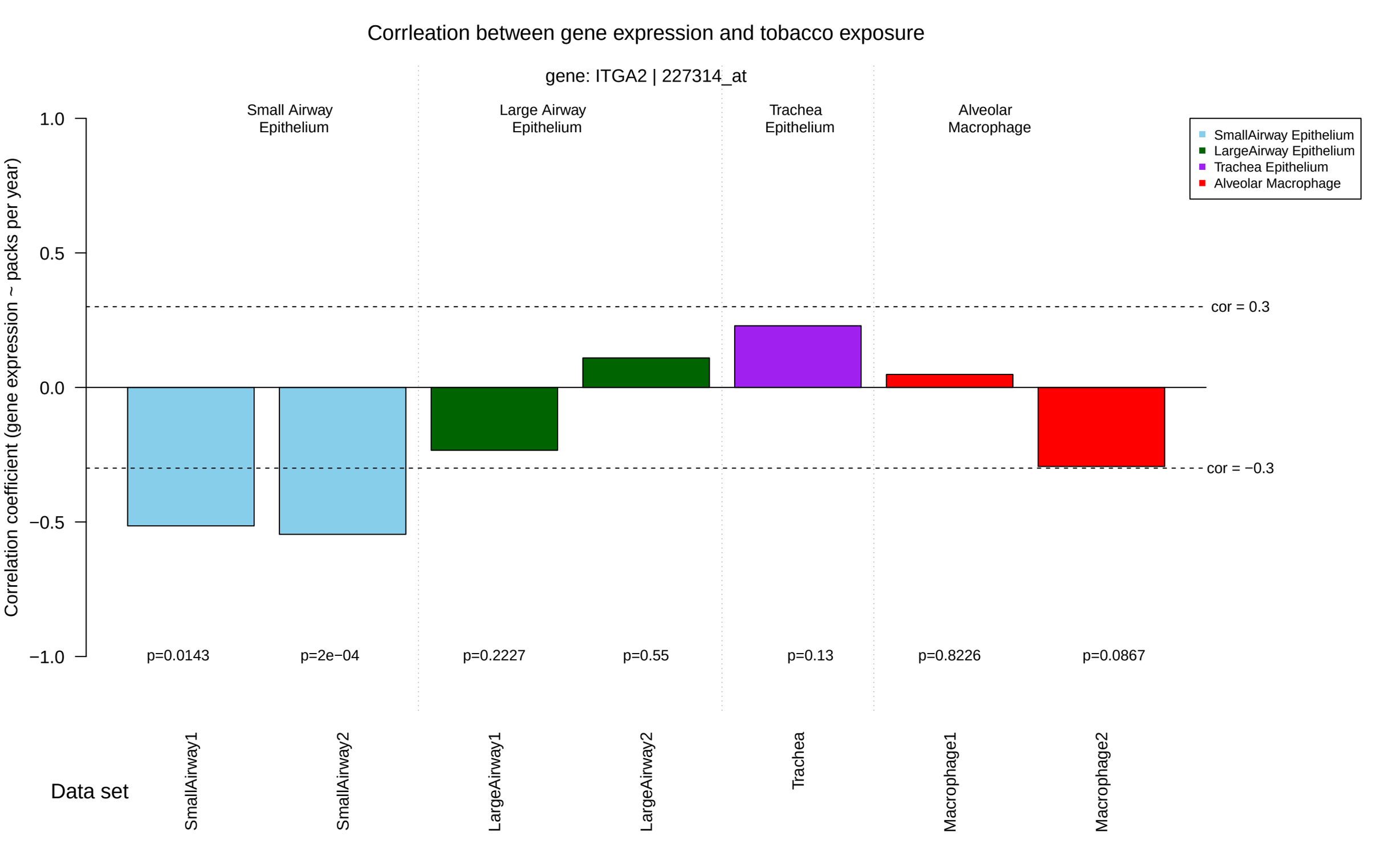

set: LargeAirway1 (p=0.0408)
ITGA2 | 227314_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7253)
ITGA2 | 227314_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0112)
ITGA2 | 227314_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8278)
ITGA2 | 227314_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1014)
ITGA2 | 227314_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ITGA2 | 227314_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0143
p=2e−04
p=0.2227
p=0.55
p=0.13
p=0.8226
p=0.0867
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2